ECN711
Galina Vereshchagina

Homework 5

1. **Numerical solution of the Bellman equation**
   Consider an infinite horizon optimal growth model with CRRA per period utility function $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ and $f(k) = Ak^{\alpha} + (1-\delta)k + y$.

   (a) Write a Matlab code which solves this problem numerically for any $\sigma \geq 0$, $\alpha \in (0,1)$ and $\delta \in [0,1]$ using the *value function iterations* algorithm described in class.

   (b) Set the following parameter values: $A = 1$, $\alpha = 0.4$, $\delta = 0.1$, $\beta = 0.96$ and $y = 0$. That is, the model is identical to the one you computed in *problem 2 of Homework 3* when $\sigma = 1$. Create a graph that would illustrate that the solutions to the model obtained by the two different approaches are the same (clearly explain what your graph shows and how it was obtained).

   (c) Explain why it is important to carefully choose the upper bound of the capital grid. Provide a numerical example demonstrating that too low upper bound leads to an inaccurate solution.

   Adjust your code so that it gives a warning if the upper bound of the capital grid is binding.

   Explain how you would find analytically the upper bound for the capital grid that certainly would not be binding.

   (d) Experiment with different number of grid points for the capital levels and compare the results and report on your observations (plotting policy functions and / or gaps between the associated value functions, measured in terms of consumption equivalence, might be useful).

   (d) Compare the policy functions under $\delta = 1$ and $\delta = 0.1$ (plot them on the same graph). Provide economic intuition.

   (e) Compare the policy functions under $y = 1$ and $y = 0.5$ (plot them on the same graph). Are your findings consistent with the analytical findings in previous homework(s)? Explain.

2. **Value function iterations with an additional state variable**
   Consider the following modification of the above model. Suppose that the fixed income y fluctuates between two states: in odd periods it is equal to y_H and in even periods it is equal to $y_L \in (0, y_H)$.

   - Set the dynamic decision problem recursively. It is convenient to denote by $V_H(k)$ and $V_L(k)$ the value functions for the odd and even periods, respectively.
   - Adjust your value function iteration code to compute the decision problem under this modification.

- Plot the policy functions for consumption and savings and analyze how they adjust as the spread between y_H and y_L increases (while maintaining the same $y_L + y_H$). Provide an intuitive explanation for the observations you make.

3. **Economy with disaster risk** (this was a qualifier problem in 2024)
Consider an infinite-horizon discrete-time economy which has access to a linear production technology $f(k) = Ak$, where $A > 0$ and k is the asset used in production. The output produced using this technology can be used for consumption or saved as an asset for the next period. In the beginning of each period, a disaster may occur (e.g., a flood, an earthquake, or a war) with probability $q \in (0, 1)$. The disaster wipes out some of the capital stock that was saved, and only fraction $\rho \in (0, 1)$ of the capital stock survives. The representative agent's preferences are given by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0, 1)$ is the time discount factor, c_t is the agent's consumption in period t, and and $u : \mathbb{R}_+ \to \mathbb{R}_+$ is a strictly increasing, strictly concave, and continuously-differentiable function, such that $\lim_{c\to 0} u'(c) = +\infty$. In period 0, the asset's value is $k_0 > 0$.

   (a) Intuitively, how would you expect optimal savings to respond to an increase in the likelihood of the disaster (i.e., higher q) or an increase in the disaster's severity (i.e., lower ρ)? Simply briefly outline the economic forces that may impact the optimal savings.

   (b) Set up recursively the social planner's decision problem in this economy.

   (c) Derive the Euler Equation and interpret it intuitively.

   (d) Suppose that $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, $\sigma > 0, \sigma \neq 1$. Derive the formula for the optimal savings rule in terms of the model's parameters.

   (e) How is the optimal savings rule affected by an increase in q? How is it affected by a decrease in ρ? Are these effects consistent with the intuition you provided in part (a) above?

4. **Setting up recursive problems**
Let's keep practicing setting up economic problems recursively. All the problems below are simple modifications of the OGM considered in class; they all are based on the the models which appeared in published research papers. Define all state and control variables and write down recursive formulation for each of the following four problems (also make sure to list all the relevant constraints):

   (a) *(Labor adjustment costs)*
   Now suppose that the labor choice is endogenous too, and the labor input can be adjusted at a cost $C(n_t, n_{t+1})$. For example, all new employees must be trained to aquire some firm-specific skill.

(b) *(Costly technology adoption)*
Now suppose that the productivity level can be either high A_H or low $A_L < A_H$ (i.e., the production technology is $A_i f(k)$, $i = L, H$). Assume that the economy starts with low productivity level A_L and can adopt a more advanced technology A_H at a fixed cost $C > 0$ at any period of time. Once the technology is adopted, it can be used forever. Of course, the planner would never want to switch back to the low-level technology, even if she were able to do it for free. First, explain intuitively why the planner would want to wait for a while before she decides to adopt the advanced technology. Then setup the problem recursively. (Remark: problems of this type are known as *discrete choice models* – because the choice variable can take only finite number of values.)

(c) Suggest an economic situation in which the agent would need to solve a dynamic decision problem (i.e., current-period action has some implications for the future) which involves a discrete choice (either once in a life-time, or in every period). Set up the agent's decision problem recursively. Carefully explain what each of the variables mean.